Farzaneh Fayazbakhsh

Biomedical Researcher | Cell Biology, Imaging & Interdisciplinary Biotechnology

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Personal Statement

Interdisciplinary biomedical researcher with Master's-level biotechnology training and current bioinformatics study, combining cell culture, molecular-biology workflows and microscopy with reproducible R/Python-based analysis. My research experience at Karolinska Institutet, KTH and SciLifeLab includes translational cell models and cross-disciplinary collaboration. I am motivated to apply this foundation to infection biophysics while developing the project's microengineering and modelling methods.

Research Experience

Karolinska Institute & SciLifeLab, Master's Thesis Research — 2026–present

- Investigating replication-stress-induced genomic instability at common fragile sites using targeted long-read sequencing.
- Performing neuroblastoma cell culture, drug treatments, DNA extraction, and long-range PCR.
- Developed bioinformatics workflows using Python, Linux and HPC environments for long-read sequencing analysis.
- Performed structural variant discovery, breakpoint characterization and endogenous barcode identification.

KTH Royal Institute of Technology & SciLifeLab, Research Engineer — 2023–2024
Sweden

- Independently developed Tumor-on-a-Chip models for in vitro anti-cancer drug screening.
- Optimized layer-by-layer cellulose nanofibril coatings to enhance cancer spheroid formation.
- Contributed to projects on Brain-on-a-Chip models for metabolic studies.
- Functionalized electrochemical sensors for ketone detection.

Tehran University of Medical Sciences, Master's Thesis Research — 2018–2022
Iran

- Synthesized gold nanoparticles conjugated with polyphenols.
- Developed microfluidic devices for endothelial cell culture.
- Modeled early-stage endothelial dysfunction and conducted oxidative stress assays.
- Assisted in preparing a PET/PDMS/Collagen nanofibrous membrane for microfluidic cell culture.

Gen-Iran Lab, Pathology and Cytology Intern — 2023
Tehran, Iran

- Hands-on experience in tissue preparation, fixation, embedding, and sectioning.
- Performed histological and immunohistochemical staining.

Wet Lab & Experimental Expertise

Cell Culture & Cancer Biology: 2D & 3D cell culture, tumor spheroid formation and organoid culture, in vitro cancer biology drug screening

Molecular Biology & Genomics Workflows: DNA extraction, downstream cellular and molecular assessments, PCR / Real-Time PCR, agarose gel electrophoresis, Agilent TapeStation analysis, flow cytometry, fluorescent microscopy

Microfluidics & Biosensors: Organ-on-a-Chip modeling, PDMS-based microfluidic systems fabrication, surface and electrochemical sensor functionalization

Histology & Pathology: Histological sample preparation, H&E staining, immunohistochemistry (IHC)

Nanotechnology & Nanomaterial Synthesis: Green synthesis of gold nanoparticles, nanofiber electrospinning, cellulose nanofibrils (CNF) coating optimization

Education

M.Sc. **Swedish University of Agricultural Sciences**, Bioinformatics — 2025–present

- Highest grade (5/5) in Bioinformatics course
- Thesis: Exploring the Potential of Fragile Sites as Markers for Lineage Tracing

KTH Royal Institute of Technology, Single-Cell & Spatial Transcriptomics Data Analysis Course — 2024

M.Sc. **Tehran University of Medical Sciences**, Medical Nanotechnology — 2017–2022
Iran
- Awarded as the Top Master's Graduate (GPA: 4/4)
- Thesis: Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip

B.Sc. **Alzahra University**, Cellular and Molecular Biology (Biotechnology Specialization) — 2012–2016
Iran

Computational Biology & Bioinformatics Skills

Programming & Pipeline Automation: Python, R, UNIX/Linux Shell, Nextflow (pipeline development), Git, and GitHub for version control

Genomics, High-Throughput Sequencing & Genetics: Practical experience in NGS data analysis, Whole Genome Sequencing (WGS), RNA-Seq processing, GWAS, QTL analysis, gene mapping, structural variation, metagenomics, genome annotation, and phylogenomics (MEGA, Clustal Omega)

Structural Biology, Modeling & Machine Learning: Sequence analysis, primer design, AlphaFold (protein structure prediction), PyMOL (molecular visualization), and foundational familiarity with deep learning frameworks

Infrastructure, HPC & DevOps: Hands-on experience with High-Performance Computing (HPC) clusters, containerization (Docker, Docker Compose, Apptainer/Singularity), and implementation of FAIR Research Data Principles

Bioinformatics & Computational Biology Projects

Automated Raw Sequencing Data Quality Control Pipeline
- Built an automated, parallelized Bash orchestration workflow designed for execution on High-Performance Computing (HPC) clusters via SLURM scheduling.
- Deployed FastQC and fastp dynamically inside isolated, version-controlled Singularity and Apptainer containers to automate raw sequence pairing, quality checking, and adapter trimming.
- Implemented multi-core processor allocation inside container runtimes to optimize throughput when executing string substitutions and batch file calculations.
- Orchestrated MultiQC container tasks to aggregate and output comprehensive data quality matrices, tracking processing adjustments across all sample cohorts simultaneously.

Tumor Microenvironment Profiling: scRNA-Seq & Spatial Transcriptomics
- Engineered end-to-end transcriptomic pipelines in R using Seurat and SCTransform to map high-dimensional gene expression across breast cancer tissue cohorts.
- Executed data pre-processing, Quality Control (QC), linear (PCA), and non-linear (t-SNE / UMAP) dimensionality reduction alongside graph-based clustering to resolve distinct cell populations and structural tissue features.
- Containerized both analysis environments using Docker and Docker Compose, ensuring full cross-platform compatibility and seamless runtime execution across varied infrastructure platforms.
- Managed pipeline documentation, software repositories, and strict workflow reproducibility utilizing Git and GitHub version control standards.

Peer-Reviewed Publications

Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip — 2023
Farzaneh Fayazbakhsh, Fatemeh Hataminia, Houra Mobaleghol Eslam, Mohammad Ajoudanian, Sharmin Kharrazi, Kazem Sharifi, Hossein Ghanbari
(Scientific Reports)

Metabolic Assessment of iPSC-Derived Neurons Under Ketogenic and Normal Diets: Ketone Sensor Development and Comparative Analysis of Neuronal Metabolism — 2026
* Both authors contributed equally to this work (co-first authors).
Rohollah Nasiri*, **Farzaneh Fayazbakhsh***, Reyhaneh Sanei, Tingting Wu, Nayere Taebnia, Rouhollah Habibey, Omid Fotouhi, John Cognetti, Volker M. Lauschke, Aman Russom, Anna Herland
(CellPress)

Assessing Layer-by-Layer Assembly of Cellulose Nanofibrils in Pancreatic Tumor Spheroid Formation 2023
Negar Abbasi Aval, Ekeram Lahchaichi, Oana Tudoran, **Farzaneh Fayazbakhsh**, Rainer Heuchel, Matthias Löhr, Torbjörn Pettersson, Aman Russom
(Biomedicines)

Characterization and Numerical Simulation of a New Microfluidic Device for Cell Investigations 2024
H Meslam, F Hataminia, H Asadi-Saghandi, **F Fayazbakhsh**, N Tabatabaei, Hossein Ghanbari
(Frontiers in Lab on a Chip Technologies)

Nanofibers in Respiratory Masks: An Alternative to Prevent Pathogen Transmission 2022
Tabatabaei SN, Faridi-Majidi R, Boroumand S, Norouz F, Rahmani M, Rezaie F, **Fayazbakhsh F**, Faridi-Majidi R
(IEEE Transactions on NanoBioscience)

Conference Presentations

Spheroid Formation in Cellulose Nanofibrils-Coated Microfluidic Chips 2024
µTAS 2024, Montreal, Canada
Poster presentation (first author)

Design of a Microfluidic Electrochemical Ketone Sensor 2023
Swedish Microfluidics in Life Science, KTH, Sweden
Oral presentation (co-author)

Investigation of Effect of Different Diets on Brain Energy Metabolism Using Organ-on-a-Chip 2023
MPS World Summit, Berlin, Germany
Poster presentation (co-author)

Preparation of PET/PDMS/Collagen Nanofibrous Membrane Embedded in a Microfluidic Device for Organ-on-Chip Applications 2022
The 3rd Conference on Nanofibers, Tehran University of Medical Sciences
Poster presentation (co-author)

Professional & Personal Skills

Bridging Wet & Dry Lab: Confident working in both experimental labs and computational environments, making it easy to translate biological questions into data-driven answers.

Independent Problem Solver: Highly autonomous in research environments, with a proven ability to troubleshoot bottlenecks and quickly teach myself new tools to keep projects moving forward.

Collaborative Team Player: Experienced in international, interdisciplinary teams such as KTH, Karolinska Institute and SciLifeLab, with a strong focus on knowledge-sharing and helping colleagues.

Clear Communicator & Fast Learner: Skilled at explaining complex data clearly through peer-reviewed publications and conference presentations; highly adaptable and quick to pick up new scientific topics and skills.

Teaching & Mentorship

KTH Royal Institute of Technology, Intern Supervisor
Sweden
- Guided an intern in laboratory techniques.

Volunteer Service, Science Teacher
- Taught science to underprivileged children.

Awards & Honors

Top Master's Graduate 2023
Tehran University of Medical Sciences

LUSH Prize Candidate 2022
Shortlisted for the Young Researcher Award.

Darolfonoon Student Award 2020
For voluntary assistance in nanofiber mask production during the COVID-19 pandemic.

Top Article Award 2014
First National and Student Conference on Applied Biology, Alzahra University, Iran
Prize winner.

Top 5% in Iran's Konkour for Master's Degree University Entrance 2016
Iran
Ranked 37th in Iran's national Konkour for the Master's Degree University Entrance Exam.

References

Dr. Kasper Karlsson
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